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PROJECT MANAGEMENT AND FINANCE

FACILITY LOCATION

WHAT IS FACILITY LOCATION?

In project management, **Facility Location** (also known as plant location) is the strategic process of determining the optimal geographical site for a company's physical operations, such as a software development center, a data center, or an office.

This decision is critical because it involves **long-term commitment and substantial capital investment** that cannot be easily reversed. A well-chosen location contributes to operational efficiency and profitability by ensuring access to customers, skilled labor, and necessary infrastructure.

KEY ASPECTS OF FACILITY LOCATION

Selecting a facility location involves three distinct levels of analysis:

A. **Selection of a Region:** Choosing a broad geographical area (e.g., a specific country or a state like Dhaka or Rajshahi).
B. **Selection of a Locality:** Identifying a specific community or district within that region.
C. **Selection of a Site:** Picking the exact plot of land or building.

WHY THESE ASPECTS ARE CRITICAL?

- **Selection of a Region:** (Selecting between Dhaka Division or Rajshahi Division)
  - **Proximity to Talent:** Does the region have top CS universities? (e.g., proximity to BUET/MIST/DU in Dhaka).
  - **Internet Infrastructure:** Availability of good ISP gateways and fiber-optic backbones.
  - **Suitability of Climate:** Cold weather reduces cooling costs for server.

WHY THESE ASPECTS ARE CRITICAL?

- **Selection of a Locality:** (Choosing between Gulshan or Kawran Bazar)

  - **Power Reliability:** Is the area on a "preferred grid" with minimal load-shedding?
  - **Communication Facilities:** High-speed internet and reliable & fast communication facilities.
  - **Local Taxes & Laws:** Incentives offered by the Bangladesh Hi-Tech Park Authority (BHTPA) such as tax holidays for 10 years.
  - **Living Conditions:** Is it easy for software engineers to commute or live nearby?

WHY THESE ASPECTS ARE CRITICAL?

- **Selection of a Site:** (Choosing a specific plot/room/floor)

  - **Price of Land/Lease:** Balancing the high cost of premium urban office space against the lower cost of suburban data center plots.
  - **Disposal of Waste (E-Waste):** IT sites must have protocols for the safe disposal of toxic electronic components, old servers, and lithium batteries.
  - **Availability for Expansion:** For a growing startup, the site must have "adequate land availability" to build additional server halls or larger "Agile pods" as the team scales.

LOCATION BREAK-EVEN ANALYSIS

The economic comparison of location alternative is facilitated by the use of cost volume profit analysis known as location break-even analysis.

The analysis can be done numerically/graphically. Following steps are involved in the break-even analysis:

a. Determine the fixed and variable costs associated with each location alternative.
b. Calculate the total cost for all locational alternatives.
c. Determine the location having the lowest total cost for the expected level of output.

LOCATION BREAK-EVEN ANALYSIS

Imagine a Dhaka-based IT firm is deciding where to host its new SaaS platform for **2,000 monthly active users.**

Location Option	Fixed Cost (Servers, Rent, Licenses)	Variable Cost (Bandwidth, Electricity, Support)
Site A: Kawran Bazar Office	Tk. 50,000	Tk. 80 / user
Site B: Kaliakair Hi-Tech Park	Tk. 80,000	Tk. 40 / user
Site C: Cloud (AWS/Azure)	Tk. 20,000	Tk. 120 / user

Total Cost Calculation= (TC = FC + VC *Volume)

LOCATION BREAK-EVEN ANALYSIS

Site A: 50,000 + (80 *2,000) = 210,000 BDT

Site B: 80,000 + (40*2,000) = 160,000 BDT (Most Economical)

Site C: 20,000 + (120 *2,000) = 260,000 BDT

Lesson: For high-volume projects, the Hi-Tech Park (Site B) becomes more profitable despite the higher upfront setup costs.

LOCATION BREAK-EVEN ANALYSIS

A, B and C are the three potential locations for producing “X” expected to sell for BDT 90. Find the most economical location for an expected volume of 1,850 units/year.

Location Option	Fixed Cost	Variable Cost
Site A	Tk. 20,000	Tk. 50 / user
Site B	Tk. 40,000	Tk. 30 / user
Site C	Tk. 80,000	Tk. 10 / user

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PRACTICE PROBLEMS: LOCATION BREAK-EVEN ANALYSIS

A software company is considering three locations to open a new development office.

Location Option	Fixed Cost (Tk./year)	Variable Cost (Tk./employee/year)
Site A: Kawran Bazar Office	Tk. 600,000	Tk. 25,000
Site B: Kaliakair Hi-Tech Park	Tk. 900,000	Tk. 18,000
Site C: Cloud (AWS/Azure)	Tk. 1,200,000	Tk. 12,000

Task01: Determine the most economical location for:

(a) 20 employees (b) 50 employees (c) 100 employees

Task02: Draw the cost volume graph.

PRACTICE PROBLEMS: LOCATION BREAK-EVEN ANALYSIS

A firm producing IoT devices is choosing a plant location.

Location Option	Fixed Cost (Tk./year)	Variable Cost (Tk./Unit)
Plant A	Tk. 500,000	Tk. 70
Plant B	Tk. 800,000	Tk. 50
Plant C	Tk. 120,000	Tk. 30

Task01: Calculate total cost for an annual output of 10,000 units.

Task02: Determine the best location for: (a) Low volume (≤ 5,000 units)(b) Medium volume (≈ 10,000 units)(c) High volume (≥ 20,000 units).

PRACTICE PROBLEMS: LOCATION BREAK-EVEN ANALYSIS

A software company wants to establish a new development center. The expected annual workload is measured in number of projects completed per year.

Location Option	Fixed Cost (Tk./year)	Variable Cost (Tk./Project)
A	Tk. 500,000	Tk. 18,000
B	Tk. 420,000	Tk. 20,000
C	Tk. 300,000	Tk. 24,000
D	Tk. 320,000	Tk. 22,000
E	Tk. 280,000	Tk. 23,000

Task01: Determine the most economical location if the expected workload is 40 projects/year. **Task02:** Draw the cost volume graph.

COST VOLUME GRAPH (SOLVED IN CLASSROOM)

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PROJECT MANAGEMENT AND FINANCE

WEEK-09|LECTURE:01

MAINTENANCE
PLANNING AND MANAGEMENT

WHAT IS MAINTENANCE?

Maintenance is the process of **keeping a system, product, software, or asset in proper working condition** after it has been developed or installed.

It includes:

- Fixing problems
- Improving performance
- Updating or replacing parts/systems
- Preventing future failures

Maintenance ensures reliability, efficiency, safety, and long-term usability.

MAINTENANCE TAXONOMY
(Based on Application Domain)

In Software Engineering:
We call it Software Maintenance

In Engineering / Project / Asset Management:
We call it Asset Maintenance or Industrial Maintenance

SOFTWARE MAINTENANCE

Software Maintenance is the process of modifying, updating, and improving software after it has been delivered, to ensure it continues to function correctly and efficiently.

Software maintenance is required to:

- Fix bugs and errors
- Adapt to new operating systems or hardware
- Improve performance
- Add new features
- Enhance security
- Minimize unplanned downtime and avoid costly disruptions.
- Ensure ongoing compliance and reduce regulatory risks

WHAT ARE THE TYPES OF SOFTWARE MAINTENANCE?

Corrective maintenance

Bug fixes, error correction, patching vulnerabilities

Adaptive maintenance

Dependency updates, configuration changes, compliance updates

Perfective maintenance

Feature enhancements, UI improvements, code cleanup

Preventive maintenance

Refactoring, removing obsolete code, documentation updates

Source: N-iX

CORRECTIVE MAINTENANCE

This type of maintenance focuses on addressing bugs, logic errors, and system faults identified after deployment.

Key activities: Debugging, deploying patches, and applying quick fixes to repair defects and restore functionality.
Trigger: Usually triggered by user reports, QA feedback, or system monitoring alerts.
Example: Resolving a login failure that occurs following the release of a new software version.

ADAPTIVE MAINTENANCE

Adaptive maintenance ensures that your software remains compatible with changes in its operating environment, such as new OS versions, third-party APIs, or updated regulations.

Key activities: Upgrading dependencies, adjusting configurations, and implementing compliance-related updates.
Trigger: Triggered by changes in platforms, technology stacks, or regulatory standards
Example: Updating cloud infrastructure integration of a specific AWS service.

PERFECTIVE MAINTENANCE

The key goal of perfective maintenance is to enhance software functionality, performance, and user experience based on evolving business needs and user feedback.

Key activities: Adding new features, optimizing algorithms, improving user interfaces, and cleaning up code for better readability and maintainability.

Trigger: Typically driven by product growth, insights from user behavior, or strategic roadmap planning.

Example: improving page load speed or redesigning a dashboard layout to enhance usability and meet updated user expectations.

PREVENTIVE MAINTENANCE

Preventive maintenance is a forward-looking approach that focuses on identifying and addressing potential issues before they impact system performance.

Key activities: Code refactoring, removing risky components, updating documentation, and improving architectural consistency.
Trigger: Typically initiated after code reviews, technical audits, or architectural assessments that reveal potential risks.
Example: Rewriting legacy modules to simplify future development work.

PRACTICE PROBLEMS

SCENARIO 1: A hospital management system is producing incorrect patient bills due to a calculation error discovered after deployment. What type of software maintenance is required?
Answer: Corrective Maintenance.

SCENARIO 2: An organization upgrades its operating system, and the existing payroll software no longer runs properly.
Answer: Adaptive Maintenance.

SCENARIO 3: Developers restructure messy and poorly written code to improve readability and reduce future bugs, even though no error currently exists
Answer: Preventive Maintenance.

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PROJECT MANAGEMENT AND FINANCE

WEEK-11

MATERIALS
MANAGEMENT

WHAT IS MATERIALS?

Materials are the physical resources used to produce goods, construct buildings, or create products.

Materials are important in project management because they directly affect project schedule, cost, quality, and risk. Proper material planning ensures timely availability, cost control, and smooth execution of project activities.

WHAT IS MATERIALS MANAGEMENT?

What is Materials Management?

Materials Management is the process of planning, organizing, purchasing, storing, and controlling materials in a way that ensures the right quality and quantity of materials are available at the right time and at the right cost.

Objectives of Materials Management:

A. Ensure continuous supply of materials
B. Minimize material cost
C. Avoid overstocking and stock-outs
D. Maintain proper quality standards
E. Reduce wastage and losses

MAIN FUNCTIONS OF MATERIALS MANAGEMENT

- **Purchasing / Procurement:** Selecting suppliers and buying materials at the best price and quality.
- **Inventory Control:** Maintaining optimal stock levels.
- **Storage / Warehousing:** Safe storage and preservation of materials.
- **Material Handling:** Efficient movement of materials inside the organization.
- **Distribution:** Supplying materials to the production department when needed.

WHAT IS MATERIALS REQUIREMENT PLANNING (MRP)

Materials Requirement Planning is a **computerized system** used to determine: **(a)** What materials are needed, **(b)** How much is needed, and **(c)** When they are needed.

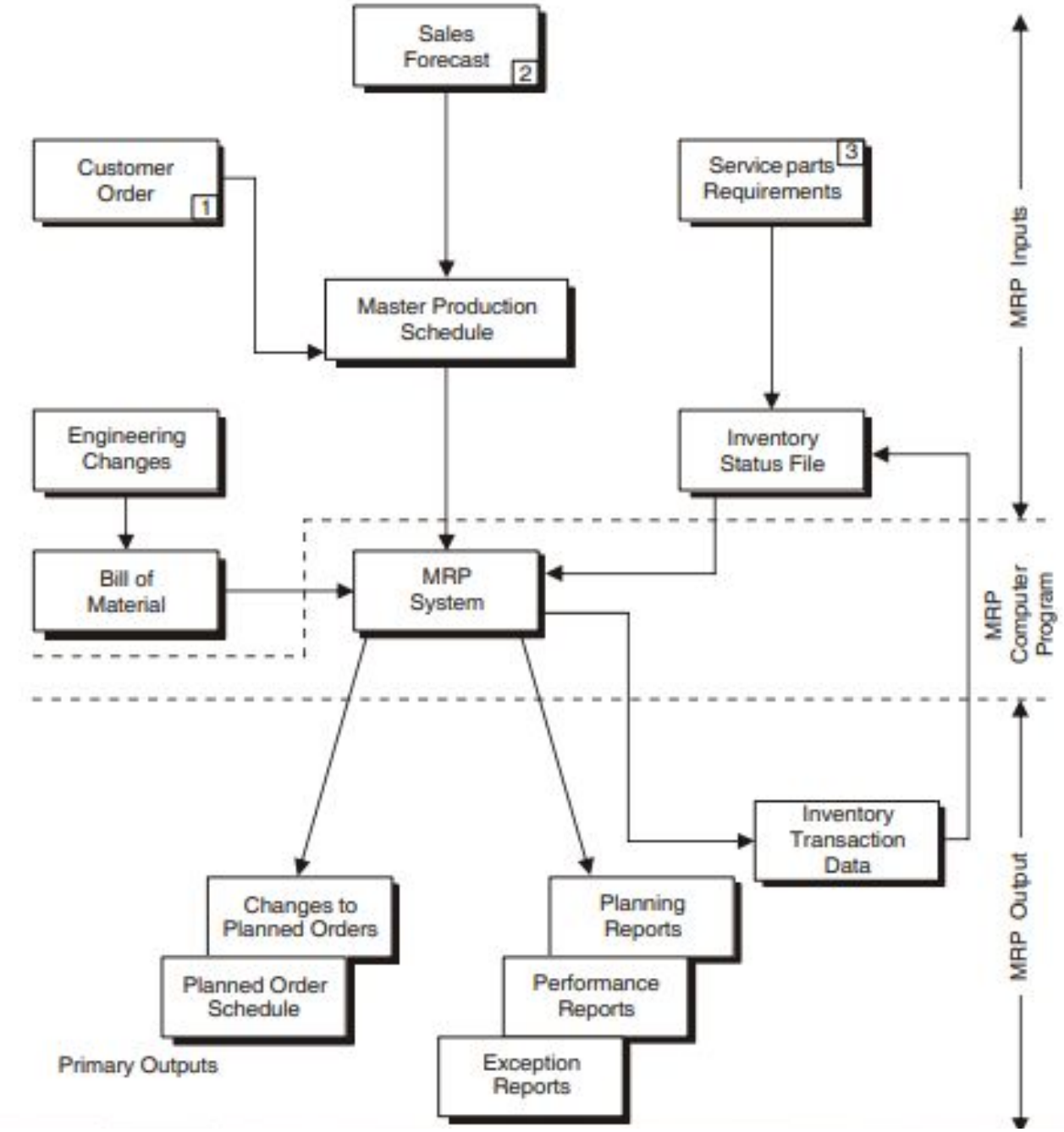
It ensures that materials are available for production at the right time while keeping inventory as low as possible.

Basically, MRP is a system that plans materials for production efficiently.

Why is MRP Needed?

A. **To Avoid Production Delays:** If materials are not available on time → production stops.MRP ensures materials arrive before they are required.
B. **To Reduce Excess Inventory:** Too much stock increases: (a) Storage cost (b) Risk of damage and (c) Capital blockage.
C. **To Improve Production Planning:** MRP connects: (a) Customer demand, (b) Production schedule, and (c) Purchasing activities.
D. **To Control Cost:** Proper planning reduces: (a)Emergency purchases, (b) Expediting cost (c) Wastage
E. **To Handle Complex Manufacturing:** In large industries (automobile, electronics), products contain thousands of parts. MRP helps manage this complexity systematically.

Materials Requirement Planning (MRP) System



MATERIALS REQUIREMENT PLANNING (MRP) SYSTEM

MRP Inputs (Top Section)

These are the essential data sources required for the MRP system to operate effectively.

A. Customer Orders:

- Confirmed product demand from customers
- Determines actual production requirement

B. Sales Forecast:

- Estimated future demand
- Helps in planning ahead

C. Master Production Schedule (MPS):

- Specifies what product to produce
- Specifies how many units to produce
- Specifies when production should take place

Bill of Materials (BOM):

- Detailed list of all components and raw materials
- Shows quantity required for each finished product

Inventory Status File:

- Current stock levels
- Safety stock information

MATERIALS REQUIREMENT PLANNING (MRP) SYSTEM
MRP Processing (Middle Section)

This is where the system analyzes and calculates material requirements.

- Determines Gross Requirements (total materials needed)
- Subtracts available inventory to find Net Requirements
- Ensures materials are available at the right time without excess stock

MATERIALS REQUIREMENT PLANNING (MRP) SYSTEM

MRP Processing (Middle Section)

These are the results generated by the MRP system for decision-making.

A. **Primary Outputs**

- Planned Order Schedule
- Changes to existing planned orders

B. **Secondary Outputs**

- Planning reports
- Performance reports
- Exception reports (alerts for shortages or delays)

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PROJECT MANAGEMENT & FINANCE

NETWORK DIAGRAM IN PROJECT MANAGEMENT

First, share your thoughts

WHAT IS A PROJECT ?

WHAT IS A PROJECT?

A project is a SET OF TASKS (TEMPORARY) done to achieve a specific goal within a limited time

Project = Temporary + Unique + Goal

SELF STUDY

- ✔ What is a network diagram in project management?
- ✔ Why is a network diagram important in project management?
- ✔ What is the critical path in project management?
- ✔ Why is the critical path important in project management?
- ✔ How does the critical path help determine the overall project duration?

LET'S IMAGINE A SCENARIO

MIST is Organizing a University Tech Fest

MIST is planning to organize a Tech Fest next month, and a group of students from CSE Dept. has been assigned to manage the entire event within a limited timeframe. Initially, the team needs 2 days to decide the event theme and finalize the date. After that, they require 3 days to prepare a budget and 2 days to get approval from the university authority. Once approval is granted, the team proceeds to book the venue (2 days) and arrange necessary logistics such as chairs, stage, and sound system, which takes 4 days. At the same time, the marketing team starts designing posters (3 days) and then continues with promotion on social media for 5 days. Meanwhile, another group works on inviting guest speakers (3 days) and confirming their participation (2 days). As promotion progresses, participant registration opens and continues for 4 days. Finally, after all preparations are completed, the Tech Fest event is executed in 1 day.

Identify all the project activities from the above scenario & arrange them in sequence.

ACTIVITIES IN SEQUENCE WITH DURATION

Activity	Description	Time
A	Decide theme & date	2 days
B	Prepare budget	3 days
C	Get approval	2 days
D	Book venue	2 days
E	Arrange logistics	4 days
F	Design posters	3 days
G	Promotion	5 days
H	Invite speakers	3 days
I	Confirm speakers	2 days
J	Registration	4 days
K	Event execution	1 day

ACTIVITY DEPENDENCY TABLE

ID	Activity	Duration (Days)	Predecessor(s)	Explanation
A	Decide theme & date	2	—	First task, no dependency
B	Prepare budget	3	A	Budget can start only after theme is decided
C	Get approval	2	B	Approval depends on budget
D	Book venue	2	C	Venue booking needs approval
E	Arrange logistics	4	D	Logistics after venue booking
F	Design posters	3	C	Poster design starts after approval
G	Promotion	5	F	Promotion after posters are ready
H	Invite speakers	3	C	Invitations sent after approval
I	Confirm speakers	2	H	Confirmation after invitation
J	Registration	4	G	Registration opens after promotion starts
K	Event execution	1	E, I, J	Final event after all preparations

NETWORK DIAGRAM

{SOLVED IN THE CLASSROOM}

MOBILE APP DEVELOPMENT

ACTIVITY	DESCRIPTION	PREDECESSOR(S)	DURATION (DAYS)
A	Requirement gathering	—	4
B	Feasibility analysis	A	3
C	UI/UX design	B	5
D	Backend development	C	8
E	Frontend development	C	7
F	Integration	D, E	4
G	Testing	F	5
H	Bug fixing	G	3
I	Deployment	H	2
J	Marketing launch	G	4

NETWORK DIAGRAM

{SOLVED IN THE CLASSROOM}

RESEARCH PROJECT

Activity	Description	Predecessor(s)	Duration (Days)
A	Define problem	—	2
B	Literature review	A	6
C	Conceptual framework	B	4
D	Questionnaire design	C	5
E	Pilot testing	D	3
F	Revise instrument	E	2
G	Data collection	F	7
H	Data cleaning	G	3
I	Data analysis	H	5
J	Report writing	I	6
K	Submission	J	2

NETWORK DIAGRAM

{SOLVED IN THE CLASSROOM}

DISASTER RELIEF PROJECT

Activity	Description	Predecessor(s)	Duration (Days)
A	Damage assessment	—	3
B	Resource mobilization	A	4
C	Recruit volunteers	A	3
D	Procure supplies	B	5
E	Transport planning	B	4
F	Distribution setup	C, E	3
G	Relief distribution	D, F	6
H	Monitoring & reporting	G	4

NETWORK DIAGRAM

{SOLVED IN THE CLASSROOM}

COMPLEX PROJECT NETWORK

Activity	Description	Predecessor(s)	Duration (Days)
A	Start	—	2
B	Task B	A	6
C	Task C	A	5
D	Task D	A	4
E	Task E	B	7
F	Task F	B, C	3
G	Task G	C	8
H	Task H	D	6
I	Task I	E, F	4
J	Task J	F, G	5
K	Task K	H, I	3
L	Task L	I, J	6
M	Task M	K, L	4
N	End	M	2

NETWORK DIAGRAM

{SOLVED IN THE CLASSROOM}